Gap-Aware Retrieval:

An Adaptive Retrieval Algorithm for Iterative Retrieval-Augmented Generation

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By

Kunal Dahal (22CSEC17)

Aniket Manger (22CSEC23)

Suman Chettri (22CSEC44)

Under the supervision of

Mr. Anmol Pandey

Assistant Professor

Computer Engineering Department



Department Of Computer Engineering

Sikkim Institute of Science and Technology

(A constituent college of Sikkim University)

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ABSTRACT

Retrieval-Augmented Generation (RAG) systems retrieve external information to support the generation of answers by large language models(LLMs). However, in complex and multi-hop questions, conventional retrieval methods may repeatedly retrieve information that is already available instead of focusing on the information that is still missing. This can limit the effectiveness of iterative retrieval.

This project is motivated by the MIGRES framework, which guides iterative retrieval using missing information. Based on the study of MIGRES and related retrieval approaches, this work proposes **Gap-Aware Retrieval (GAR)**, a retrieval algorithm designed to select documents according to the current information gap and previously collected evidence. GAR considers three factors: gap relevance, gap coverage, and evidence novelty.

The proposed work aims to investigate whether explicitly incorporating the current information gap and previously collected evidence into the retrieval process can lead to more useful evidence selection in iterative RAG. The proposed GAR algorithm will be implemented within the MIGRES framework and evaluated against established retrieval approaches.

Keywords: Retrieval-Augmented Generation, Information Retrieval, MIGRES, Gap-Aware Retrieval, Information Gap, Multi-Hop Question Answering.

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CHAPTER 1

INTRODUCTION

Retrieval-Augmented Generation (RAG) combines information retrieval with large language models (LLMs) to improve answer generation using information from external documents. Instead of relying only on the knowledge stored in the model, RAG retrieves relevant information and provides it as context to the LLM. The quality of the retrieved information therefore has an important effect on the final answer.

Traditional retrieval methods mainly select documents based on their relevance to the given query. This can become challenging for complex and multi-hop questions, where the answer may require information from multiple documents. After one retrieval step, some information may already be available while other information is still missing. In such cases, retrieving documents based only on the original query may result in repeated or less useful information.

The MIGRES framework addresses this issue by using missing information to guide iterative retrieval. After recovering and processing the evidence, MIGRES identifies what information is still missing and uses it to guide the next retrieval step [1]. This provides a useful direction for improving retrieval in multi-step question answering.

Recent research has also explored other ways to improve the usefulness of the retrieved information. Relevant Information Gain considers whether retrieved information provides useful new content rather than unnecessary repetition [2]. InfoGain-RAG similarly investigates document information gain for reranking and filtering retrieved documents [3]. Other work has considered source reliability as an additional factor in RAG [4], while ColBERTv2 provides an efficient late-interaction approach for document retrieval [5].

From the study of MIGRES and these related approaches, this project identified a possible opportunity to improve iterative retrieval. The main observation is that the retrieval process could consider not only the original question but also what information is currently missing and what evidence has already been collected.

The study of MIGRES and related retrieval approaches has motivated further investigation into how iterative retrieval can be improved. This project will explore the idea of **Gap-Aware Retrieval (GAR)**, focusing on whether the current information gap and previously collected evidence can help guide iterative retrieval. The concept will be further studied, refined, and developed during the subsequent stages of the project.

CHAPTER 2

LITERATURE SURVEY

Table 2.1: Literature Survey 1

Name of paper	**LLMs Know What They Need: Leveraging a Missing Information Guided Framework to Empower Retrieval-Augmented Generation**
Authors	K. Wang, F. Duan, P. Li, S. Wang, and X. Cai
Publication Details	Proceedings of the 31st International Conference on Computational Linguistics (COLING), 2025, pp. 2379–2400.
Research Findings	• MIGRES is introduced as a framework for iterative Retrieval-Augmented Generation. • Missing information is identified after each retrieval step and is used to guide the next retrieval. • The framework shows the importance of identifying what information is still required during iterative retrieval.
Research Gap	• The retrieval process is guided by missing information, but further investigation can be made into how the retrieved evidence itself can be considered during document selection.
Relevance with Our Work	• MIGRES is the main starting point for our project. • The study of MIGRES motivated the investigation of improving the retrieval stage of an iterative RAG system.

Table 2.2: Literature Survey 2

Name of paper	**Better RAG using Relevant Information Gain**
Authors	M. Pickett, J. Hartman, A. K. Bhowmick, R. Alam, and A. Vempaty
Publication Details	arXiv preprint, 2024.
Research Findings	• The use of information gain is investigated as an additional factor for improving RAG retrieval. • Retrieved information is considered not only for its relevance, but also for the useful new information it can provide. • The approach aims to reduce unnecessary repetition in retrieved information.
Research Gap	• The usefulness of new information is considered, but the information requirement can change after each step in an iterative retrieval process.
Relevance with Our Work	• The paper provides motivation for reducing repeated information during retrieval. • The idea of considering useful new information is relevant to our investigation of iterative retrieval.

Table 2.3: Literature Survey 3

Name of paper	**InfoGain-RAG: Boosting Retrieval-Augmented Generation through Document Information Gain-based Reranking and Filtering**
Authors	Z. Wang et al.
Publication Details	Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025, pp. 7190–7204.
Research Findings	• Document information gain is studied as a way to improve RAG retrieval. • Reranking and filtering are used to give greater importance to documents that provide useful information. • The work emphasizes that document usefulness should not be based only on relevance.
Research Gap	• The focus is placed on information contribution during document selection. • Further investigation can be made into using information contribution together with the changing information requirement in iterative retrieval.
Relevance with Our Work	• The paper supports the consideration of information contribution during document retrieval. • This provides motivation for investigating whether useful new evidence can be selected based on the current information requirement.

Table 2.4: Literature Survey 4

Name of paper	**Retrieval-Augmented Generation with Estimation of Source Reliability**
Authors	J. Hwang, J. Park, H. Park, D. Kim, S. Park, and J. Ok
Publication Details	Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025, pp. 34279–34303.
Research Findings	• The reliability of information sources used in RAG is investigated. • Source reliability is considered as an additional factor when selecting information for RAG. • The study shows that the quality of retrieved sources can affect the final generated answer.
Research Gap	• Source reliability provides information about the quality of a source, but it does not directly indicate whether the source contains the information currently required by an iterative retrieval process.
Relevance with Our Work	• The paper shows that factors other than query relevance can be considered during RAG retrieval. • This supports the investigation of multiple factors for improving evidence selection.

Table 2.5: Literature Survey 5

Name of paper	**ColBERTv2: Effective and Efficient Retrieval via Lightweight Late Interaction**
Authors	T. Santhanam et al.
Publication Details	Proceedings of NAACL, 2022.
Research Findings	• ColBERTv2 introduces an efficient late-interaction approach for document retrieval. • Query and document representations are compared at the token level to improve document matching.
Research Gap	• The retrieval approach focuses on matching queries with documents. • The information already collected and the information still missing during iterative retrieval are not the main focus of the method.
Relevance with Our Work	• ColBERTv2 provides an established retrieval method that can be used for comparison. • It also provides a different retrieval approach from the information-gap-based direction being investigated in this project.

CHAPTER 3

PROBLEM DEFINITION

1. In iterative RAG, conventional retrieval may repeatedly retrieve documents that are relevant to the original query but contain information that has already been obtained. This can lead to redundant retrieval and reduce the usefulness of later retrieval steps.
2. The information required to answer a multi-hop question can change as new evidence is collected. Existing retrieval methods may not sufficiently use the current missing information when selecting documents for the next retrieval step.
3. Previously retrieved evidence can contain information that is already available to the system. A retrieval mechanism is therefore needed to consider the collected evidence along with the remaining information requirement when selecting further documents.
4. The effectiveness of using the current information gap and previously collected evidence for iterative retrieval needs to be experimentally investigated and compared with existing retrieval approaches.

CHAPTER 4

OBJECTIVES

1. To investigate methods for reducing redundant retrieval of information that has already been obtained.
2. To study and incorporate the current missing information as a factor in selecting documents during iterative retrieval.
3. To investigate how previously collected evidence can be considered when selecting documents for subsequent retrieval steps.
4. To evaluate the effectiveness of the developed retrieval approach against existing retrieval methods using suitable retrieval and question-answering measures.

CHAPTER 5

APPLICATIONS

1. **Multi-Hop Question Answering:** The approach can be investigated for question-answering tasks that require information from multiple documents or retrieval steps.

2. **Retrieval-Augmented Generation:** The approach can be studied as a retrieval component for iterative RAG systems where additional information is retrieved as the answer is developed.

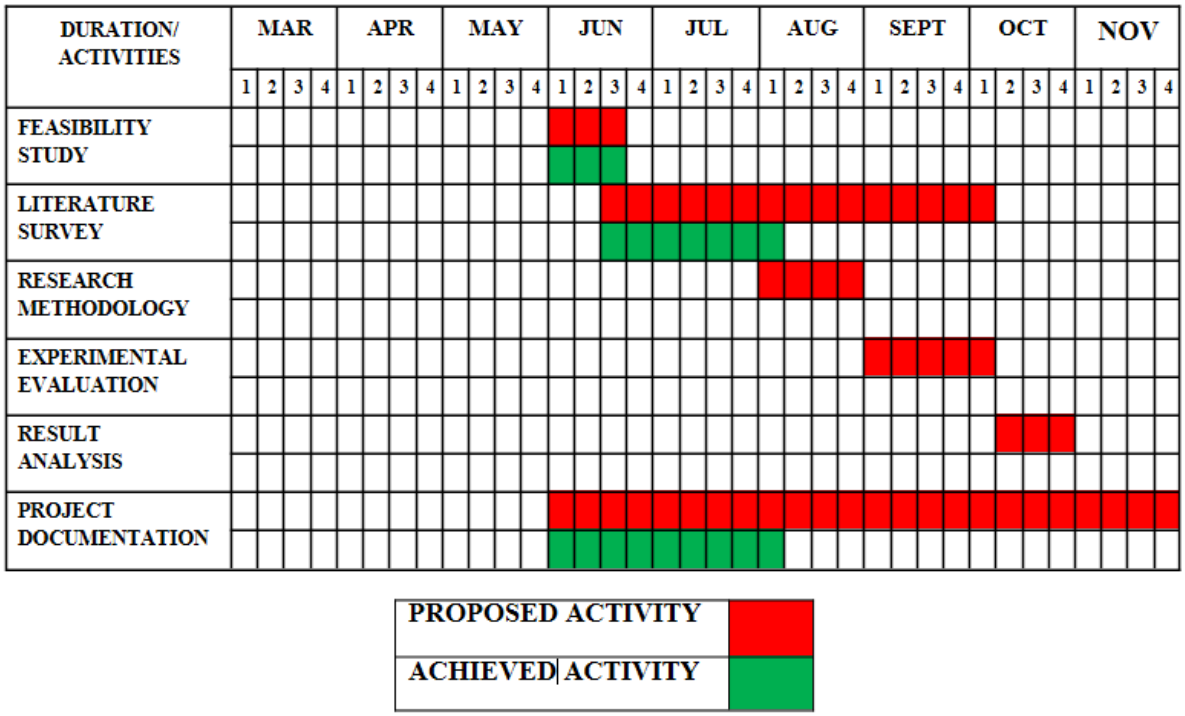
3. **Research and Information Search:** The approach may be useful for research-oriented systems where information is collected from multiple documents and previously retrieved evidence needs to be considered.

4. **Evidence-Guided AI Systems:** The approach may be applicable to AI systems that repeatedly retrieve external information and need to focus subsequent searches on information that is still required.

CHAPTER 6

PLANNING

Figure 6.1: Gantt Chart



CHAPTER 7

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